

ARIA — Project Timeline

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By 3 August 2026

- The 5-DoF arm should be modeled to match the SG90/MG995 joint layout and placed within a simulated workspace, with a table surface and a bin of test objects included in the scene.
- Objects should be spawned across a range of positions, orientations, and sizes to create a representative test set for the perception pipeline.
- The simulated camera and IMU sensors should be brought online, and the YOLO-based object-detection pipeline should be validated against the spawned objects.
- Baseline detection accuracy and frame rate should be recorded to establish a reference point for later comparisons as the pipeline grows more complex.

By 10 August 2026

- Z-coordinate recovery should be implemented using camera and table-height calibration geometry, without relying on any dedicated depth sensor.
- Recovered coordinates should be validated against the simulator's ground-truth object poses across the full test object set, and the calibration should be tuned until error falls within an acceptable margin.
- Edge cases such as objects near the boundary of the workspace or partially out of frame should be tested to understand where the calibration approach breaks down.
- A short internal report should document detection and localization accuracy at this stage, to be carried forward into grasp planning.

By 17 August 2026

- Depth-Anything and MiDaS should be benchmarked on rendered frames under varied simulated lighting, and the better-performing model should be selected for the pipeline.
- SAM2 should be integrated for instance segmentation, and its output should be compared against plain bounding-box detection for grasp-region accuracy.
- Grasp candidates should be generated for the first batch of simulated objects using GraspNet and AnyGrasp, and their outputs should be compared on candidate quality and generation speed.
- The Affordance Agent should filter and rank these candidates by how each object is actually used, rather than by the nearest reachable surface.

By 24 August 2026

- The available IK solvers, ikpy, roboticstoolbox-python, trac-ik, KDL, and an analytical solution, should be benchmarked on the arm's specific geometry for accuracy, solve time, and singularity handling.
- The best-performing solver should be selected and integrated with MoveIt2 for trajectory execution.

- The first successful simulated trajectory should be demonstrated: the arm reaching, grasping, and lifting a target object without collision, using cubic or quintic spline trajectories.
- Collision-checking and cable-aware planning constraints should be verified against the simulated workspace, including scenarios designed to stress-test the planner.

By 31 August 2026

- The Planning Agent, built on Llama 3.1 8B served through Ollama, should be integrated to decompose spoken or typed commands into a command, goal, subgoal, and action hierarchy.
- Function-calling should be wired so the Planning Agent can dispatch work to the correct specialist agent, including the Skill, Reachability, and Affordance agents.
- Tree-of-Thought or Monte Carlo Tree Search exploration should be added for tasks with genuine branching complexity, and tested against a simpler direct-planning baseline.
- Affordance-based grasp selection should be demonstrated end-to-end on simulated objects such as cups and tools, closing the loop from language command to selected grasp.

By 7 September 2026

- Per-action confidence scoring should be implemented, with the system pausing to request user approval whenever confidence falls below a defined threshold.
- The World Model Agent should maintain a running semantic map of object coordinates, color and material attributes, and lifecycle state across a simulated session.
- Scenarios where an object is not where the map expects it to be should be tested, confirming the system asks the user rather than assuming a detection failure.
- The Safety Agent should be validated against deliberately constructed collision-risk and cable-entanglement scenarios in simulation.

By 14 September 2026

- Performance metrics, pick success rate, IK error, localization error, planning time, and inference latency, should be logged across repeated simulated trials.
- The Learning Agent should complete its first improvement pass using this logged data, targeting measurable gains in pick success rate over the baseline recorded in earlier weeks.
- The Evaluation Agent should classify failures into categories such as missed object, object slipped, IK failure, collision, and perception error, to give the results structure.
- End-to-end pick, place, stack, and sort tasks should be completed across a fixed set of simulated test scenes, with results compared against the categories above.

By 21 September 2026

- The full pipeline should be integrated end-to-end: a spoken or typed command parsed, the target object located, a grasp planned and executed, and the outcome verified.
- The chain-of-thought logging output should be finalized so every decision the system makes is traceable and presentable.

- Bugs surfaced during rehearsal runs should be resolved, and the system's behavior should be re-validated across the full test scene set one final time.
- The recorded simulation demonstration and supporting materials should be finalized for submission.